COMMODITY INVESTMENT FORMS

Learning Outcome

- describe features of commodities and their investment characteristics

Commodity Investment Features

Commodities themselves do not generate cash flows but usually incur costs (*cost of carry* introduced in derivatives learning modules), such as those for transportation, storage, and insurance for physical commodities. Investors seek to benefit from commodity price appreciation (in excess of carry cost) based on their future economic value rather than actual use of the underlying asset.

Moreover, governments have realized the importance of controlling natural resources and taken an increasingly important role in natural resource markets. For instance, many governments provide food price subsidies to customers and price support to farmers.

Governments often control extractable natural rights, such as energy resources and mining. In many emerging markets, governments or government-owned enterprises control strategic energy production or mining resources. For instance, SOCAR, the State Oil Company of Azerbaijan Republic, is the fully state-owned national oil and gas company of the country and extracts oil and natural gas from onshore and offshore fields of the Caspian Sea. Moreover, it operates Azerbaijan's only oil refinery and operates several oil and gas export pipelines. It is a major source of income for the country. In other countries, owners of land may only be able to cultivate the soil and may extract only certain minerals. Often the government owns and manages subsurface rights and has the right to extract certain resources, such as oil, gas, coal, gold, and silver.

Environmental factors play a direct role in natural resource investments, because governments are increasingly implementing environmental safeguards to meet climate objectives and control activities with climate impact, such as mining, agriculture, and energy extraction and production. More specifically, global climate change policies seek to reduce reliance on fossil fuels (coal, oil) and increase renewable energy use. To do so, countries are in the process of adopting national programs that intend to increase renewable energy (wind, solar, biomass) and reduce the reliance on fossil fuels.

The policy objective of reducing the reliance on fossil fuels has shifted focus to electric vehicles and advances in battery technology. A potential impact of the reliance on low-carbon energy technologies is the higher demand for many minerals and metals, such as lithium, cobalt, and nickel. As mining activities for these critical metals will increase, there will be significant impacts on local water systems, ecosystems, and communities.

Finally, there is growing interest from ESG investors seeking to promote sustainable farming practices or use timberland investments for carbon offsets.

Distinguishing Characteristics of Commodity Investments

Commodity sectors include precious and base (i.e., industrial) metals, energy products, and agricultural products. Exhibit 2 offers examples of each type. The relative importance, amount, and price of individual commodities evolve with society's preferences and needs. Increasing industrialization of emerging markets has driven strong global demand for commodities. These markets need increasing amounts of oil, steel, and other materials to support manufacturing, infrastructure development, and the consumption demands of their populations. Emerging technologies, such as advanced cell phones and electric vehicles,

Rate Your Confidence
High
Medium
Low
Continue »

Category
Natural Resources

Exhibit 2:

Examples of Commodities

Sector	Sample Commodities
Energy	Oil, natural gas, electricity, coal
Base metals	Copper, aluminum, zinc, lead, tin, nickel
Precious metals	Gold, silver, platinum
Agriculture	Grains, livestock, coffee
Other	Carbon credits, freight, forest products

Commodities may be further classified by physical location and grade or quality. For example, there are many grades and delivery locations for crude oil and wheat. Commodity derivative contracts thus specify quantity, quality, maturity date, and delivery location.

The majority of commodity investing is implemented through derivatives. Physical commodities often generate unwelcome tax obligations and costs arising from storage, insurance, brokerage, and transportation. Additionally, physical commodity markets lack price transparency. As such, commodity investments are usually made through financial derivative instruments, most frequently commodity futures and forwards and occasionally options on futures. Using derivatives to establish exposures to natural resources has several benefits: Because these instruments are traded on organized exchanges, they are very liquid and provide opportunities for price discovery.

Futures contracts are obligations to buy or sell a specific amount of a given commodity at a fixed price, location, and date in the future. Futures contracts are exchange traded, are marked to market daily, and may or may not be settled on delivery or receipt of the physical commodity at the end of the contract. This delivery obligation becomes dramatically important during stressful periods. For example, with oil during the global financial crisis in 2008 and during the COVID-19 pandemic in 2020, as demand collapsed, oil producers could not find buyers and global storage filled suddenly. Even commodity-related ETFs were affected, forcing some to close and impose large losses on investors.

For futures contracts, counterparty risk is managed through the settlement process between the clearinghouse/exchange and clearing brokers. Commodity exposure can be achieved through means other than direct investment in commodities or commodity derivatives, including the following:

- *Exchange-traded products.* ETPs, either funds (ETFs) or notes (ETNs), may be suitable for investors restricted to equity shares or seeking simplified trading through a standard brokerage account. ETPs may invest in commodities or commodity futures. For example, the SPDR Gold Shares ETF seeks to track the price of physical gold by holding bullion in vaults. It owned just under USD53 billion in gold bullion as of December 2022. ETPs may use leverage and may replicate the pay-offs from a long or short position, with the latter form considered inverse or “bearish.” Similar to mutual funds or unit trusts, ETPs charge fees included in their expense ratios.
- *Investing with commodity trading advisers.* CTAs are another way to gain commodity exposure. CTAs are managed futures funds that make directional investments primarily in futures markets based on technical and fundamental strategies. A commodity-focused CTA might concentrate on a specific commodity (such as grains) or be broadly diversified across commodities. However, one would need to find a fund focused solely on the desired commodity, because modern CTAs often invest in a variety of futures, including commodities, equities, fixed income, and foreign exchange. Individual investors may establish accounts that are managed in accordance with their specific investment preferences and risk tolerance called **separately managed accounts**

hedge funds.

- *Specialized funds investing in specific commodity sectors.* An example of specialized funds is private energy partnerships, which are similar in structure to private equity funds and enable institutional exposure to the energy sector. Management fees can range from 1% to 3% of committed capital, with a typical life span of 10 years and extensions of 1- and 2-year periods. Publicly available energy mutual funds and unit trusts typically focus on the oil and gas sector, often acting as fixed-income investments to pay dividends from rents or capital gains. They may focus on upstream (drilling), midstream (refineries), or downstream (chemicals). Their management fees are comparable with those of other public equity managers and range from 0.4% to 1%.

Commodity investments are typically direct underwriting of the acquisition, management, and extraction of the natural resource itself. These investments usually involve direct or indirect claims to *physical* assets themselves and not claims on residual (equity) or fixed (debt) cash flows.

This indirect approach, however, comes with the added risks from financing and operations and, when privately held, the impact of additional illiquidity.

Basics of Commodity Pricing

Investors seeking to benefit from direct commodity price exposure typically use derivative instruments, such as exchange-traded futures and options and forwards, with individual commodities or an index as the underlying asset. Different commodity indexes are composed of different commodities and have materially different index weights, which determines varying exposures to not only specific commodities but also commodity sectors. However, as we will see later in this module, the correlation between commodities and traditional asset classes is typically low, which means improved portfolio diversification is possible regardless of the index chosen.

Since commodities trade in both physical and financial markets, there is a direct relationship between their prices in both the physical cash markets and the financial derivative markets. Such no-arbitrage conditions, which we have encountered earlier in the derivatives readings, dictate that the difference in prices between the cash or spot markets and derivative markets is equal to the *cost of carry*, which is the opportunity cost of holding these assets, and mirrors the risk-free rate and the cost of storing, transporting, and insuring the commodity. The holder of the physical commodity should expect to be compensated via a higher forward price, $F_0^+(T)$, than the prevailing cash price.

However, there may be non-cash benefits from holding the physical commodity instead of gaining exposure using a derivative. Such *convenience yield* may arise under conditions when the market participants prefer to hold the physical commodity; for instance, the owners want to ensure that they have continuous access to this commodity. Usually, the convenience yield is related inversely to inventory levels of the underlying commodity. Because convenience yield is a benefit and *accrues* to the owner, it *reduces* the forward price.

The pricing relationship between cash, S_0, and derivative markets can be expressed under continuous compounding as

$$F_0(T) = S_0 e^{(r+c-i)T}, \qquad 1$$

where c is the cost of carry, i is the convenience yield, r is the risk-free rate, and T is the time to the expiration of the forward contract. Effectively, the relationship between commodity forward and spot prices over time depends on the relative relationship between the cost of carry and the convenience yield.

Looking at the relationship between the convenience yield and the cost of carry, there are two relationships. When the spot price is above the forward prices, there is **backwardation**, a downward-sloping, or inverted, forward curve. This can occur for

forward prices, there is **contango** because the cost of ownership exceeds the benefit of a convenience yield and the forward price will be above the underlying spot asset price. As a rule of thumb, a contango scenario generally lowers the return of the long-only investor, and a backwardation scenario enhances it. The following example shows how market changes can lead to significant shifts in the shape of the commodity forward curve:

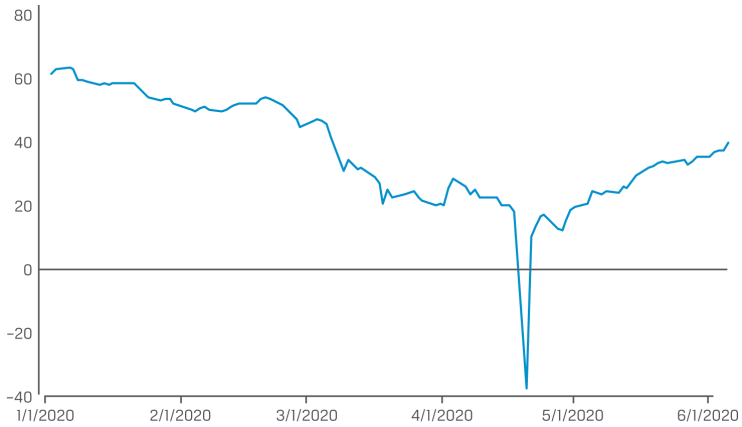
EXAMPLE 1

Crude Oil Going from Backwardation to Contango

In April 2020, the price of crude oil futures on the New York Mercantile Exchange (NYMEX) fell below zero for the first time ever; sellers paid buyers to take on an exposure to oil. This situation is highly unusual and was caused by the lockdowns in the wake of the COVID-19 pandemic that eroded the demand for oil. Producers could not cut crude oil production quickly enough, and storage facilities were overflowing with oil. This caused oil inventories to skyrocket. Exhibit 3 shows the NYMEX oil futures price from January to June 2020.

Exhibit 3:

NYMEX Oil Futures Price (US dollars per contract), January–June 2020



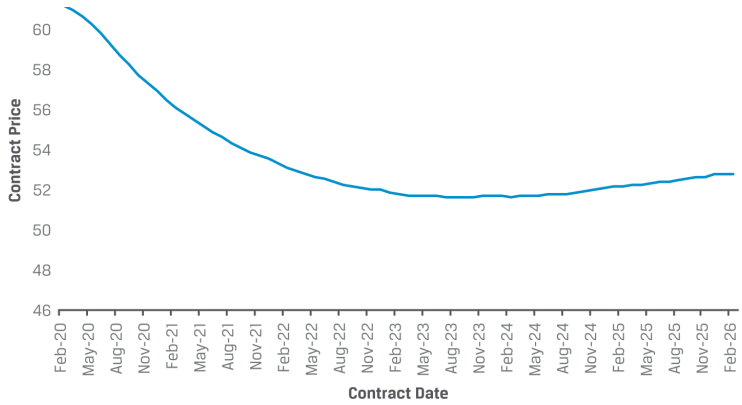
Source: Bloomberg.

The sudden disappearance of the demand for oil impacted the shape of the oil futures curve, the series of oil futures contract prices with different forward settlement dates. Prior to the outbreak of the COVID-19 pandemic in January 2020, the US oil markets had the lowest inventory levels in over a year and expected a continued healthy demand from a vibrant economy. Because of low inventory levels, the oil futures curve was in backwardation at the beginning of January 2020 because the benefit of holding oil outright exceeded the cost of carry of having oil in inventory (see Exhibit 4).

Exhibit 4:

NYMEX Oil Futures Curve, January 2020

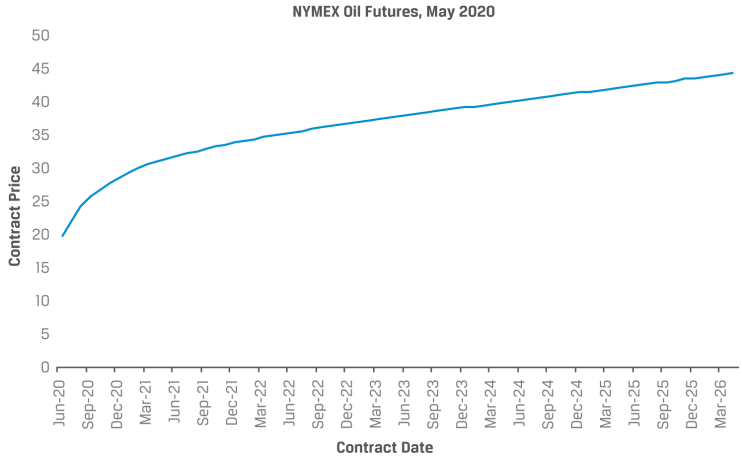
Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search



As lockdowns continued into mid-2020, US oil market inventory approached near all-time high levels but demand remained muted. Carry costs then outweighed any benefit of storing oil for production purposes (and there was hardly any open storage capacity available), and the oil futures curve was in *contango* at the beginning of May 2020 (see Exhibit 5).

Exhibit 5:

NYMEX Oil Futures Curve, May 2020



Question Set

1. Compare farmland investment to commodity investment with respect to sources of value.

   Hide Solution

   Solution:

   A similarity between the two is that both forms of investment rely on changes in the price of underlying. A difference between the two is that commodity investments, which typically rely on derivative contracts, do not generate cash flows while farmland investment generates cash flows through the income generated from harvesting the agricultural crops.

2. Which of the following statements is most correct about investors seeking commodity exposure through a commodity trading adviser (CTA)?

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

C. The investor is seeking to profit from specific directional trends in commodity futures contracts.

Hide Solution

Solution:

C is correct. Commodity trading advisers devise trading strategies using derivative contracts on commodities that are focused on predicting upcoming bull or bear trends. A is incorrect because CTAs do not advise on physical commodity transactions. B is incorrect for a similar reason, because the income stream from most commodities would require direct or indirect land ownership, which produces commodities.

3. Discuss how the relationship between costs of carry and benefits of owning a commodity outright affect the relationship between the forward price and spot price of the commodity.

Hide Solution

Solution:

A simple framework for expressing the relationship between forward price and spot price in terms of costs and benefits of owning the underlying commodity is as follows:

Forward price = Spot price + Costs of carry – Benefits of ownership.

Thus, if costs of carry exceed benefits of ownership, then forward price is greater than spot price for the commodity. In contrast, forward price is less than spot price if benefits exceed costs.

4. Explain why low inventories of a commodity may result in backwardation for the commodity.

Hide Solution

Solution:

Backwardation reflects a downward-sloping term structure of prices for a commodity. In the near term, backwardation implies forward price below spot price. Low inventories of a commodity cause investors to prefer to hold the physical commodity over derivative contracts (i.e., forwards). This preference will cause price for the physical asset to be bid higher, and if this non-cash benefit of owning the physical commodity (also known as convenience yield) exceeds costs of ownership of the commodity (such as interest and storage), then the spot price of the commodity exceeds its forward price.

Discuss Filter

Discussion

What do you want to discuss... (required)

NA backwardation et contango refer to forward price or futures? Definition is confusing

Created 4 months ago by Nada Ait labsir 2 replies | Last Activity: 4 months ago

Show All replies Reply to this Comment

© 2025 CFA Institute. All Rights Reserved.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search